

Fig. 3: Actual-size PCB layout of the electronic door lock system

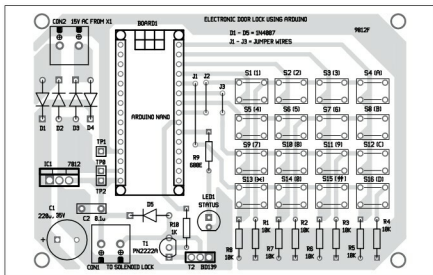


Fig. 4: Component layout of the PCB

Software

The software code (door lock.ino) for Arduino Nano is written in Arduino programming language and compiled using Arduino IDE.

Keypad.h header file is added to the library for the functioning of the keypad. This library can be downloaded from <http://playground.arduino.cc/code/Keypad#Download>

After downloading, import this library to Arduino IDE from Sketch-> Import Library...-> Add Library option.

The default password is *1234* that can be modified by making changes to the code in the following line:

```
char PIN[4] = {'1','2','3','4'}
```

Construction and testing

Any Arduino-Nano- or UNO-compatible board can be used for this project. However, Arduino Nano is recommended as it is small and compact. It accepts 6V DC to 20V DC external power supply. Use of a 230V AC primary to 15V, 1A secondary transformer (X1) is recommended. Here we have used a regulated supply circuitry using 12V regulator IC 7812 to drive the circuit and solenoid lock.

An actual-size, single-side PCB layout of the electronic door lock system is shown in Fig. 3 and its component layout in Fig. 4. You can also use a readymade keypad module in place

PARTS LIST

Semiconductors:

- BOARD1 - Arduino Nano board
- IC1 - 7812, 12V regulator
- LED1 - 5mm LED
- T1 - PN2222A npn transistor
- T2 - BD139 npn transistor
- D1-D5 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R8 - 10-kilo-ohm
- R9 - 680-ohm
- R10 - 1-kilo-ohm

Capacitors:

- C1 - 220 μ F, 35V electrolytic
- C2 - 0.1 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 15V, 1A secondary transformer
- S1-S16 - 4x4 matrix keypad using tactile switches
- CON1, CON2 - 2-pin connector terminal - 12V solenoid lock

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

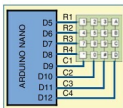


Fig. 5: 4x4 matrix keypad

of the switches S1 through S16 given in this PCB. Readymade keypad interfacing with Arduino is shown in Fig. 5.

Connect transformer X1 to the PCB at CON2 (Fig. 4). Initially, the solenoid lock (connected on CON1) will be locked and the status LED (LED1) connected on digital pin D3 of Arduino will be off.

When the user enters the right password, the solenoid lock gets unlocked for five seconds and LED1 glows. After five seconds, both LED1 and solenoid lock will be in the initial off state.

If the password is incorrect, LED1 will blink a few times, indicating that a wrong password has been entered.

The different voltage levels of 12V (TP2) and 5V (TP1) can be measured with respect to 0V (TP0), as listed in the table. ●



Joy Mukherjee is an electronics hobbyist and a small-business owner in Albany, New York, the USA. His interests include designing radio frequency (RF) circuits